

Ignorance, lexical domain restriction and the Mandarin *wh*-numeral *jǐ*

Minghui QIU^{1,2}

Boston University¹, Zhejiang University², qmh@bu.edu

Introduction

Background Mandarin *wh* words can be used existentially as indefinites, besides their interrogative use in questions.

- Many previous accounts regard them as Negative Polarity Items due to their limited distribution under the indefinite use, e.g. in Downward Entailment environments (Huang 1982, Cheng 1991, a.o).
- Liu & Yang (2021) observe that they can be licensed in positive environments, but crucially, with **ignorance inference** obligatorily triggered.

- (1) *Zhāngsān cóng shéi nàr jiè-le běn shū*
Zhāngsān from who/someone there borrow-PFV CL book
Interrogative: ‘Whom did Zhāngsān borrowed a book from?’
Existential: ‘Zhāngsān borrowed a book from someone (that I don’t know).’

New Observation The *wh*-numeral *jǐ* ‘how many/several’ can also be used both in interrogatives and existentials.

- However, the latter use occurs freely with no distributional limitation observed as other *wh*-items;
- and, more crucially, no (obligatory) ignorance inference is triggered under the existential use.

- (2) *Zhāngsān jiè-le jǐ běn shū*
Zhāngsān borrow-PFV how.many/several CL book
Interrogative: ‘How many books did Zhāngsān borrow?’
Existential: ‘Zhāngsān borrowed several books.’
☞ Can be followed by *jùtǐ lái shuō shì liù běn* (‘specifically, it was six in number’).

Questions

- Are the interrogative and existential *jǐ*’s the same? If so, what is its semantics, and how are the different readings derived?
- Why is *jǐ* not associated with an obligatory ignorance inference under the existential use?

Proposal

- The interrogative and existential *jǐ*’s share the same semantics. It denotes a restricted Hamblin set of numbers from 1 to 9, but not all numbers, which we regard as a lexical domain restriction.
- The different readings are derived from the association of *jǐ* with existential or question operators.
- Jǐ* has no obligatory ignorance effect since its lexically restricted domain makes it informative in its existential use.

Analysis (Part I: *jǐ* denotes 1 to 9)

Gist: *Jǐ* is a simplex numeral; accordingly, it denotes numbers from 1 to 9, exactly the range of numbers denoted by non-*wh* simplex numerals.

Background on Numerals: Mandarin numerals can be divided into simplex and complex ones.

- Simplex numerals consist of *yī* ‘1’ to *jiǔ* ‘9’.
- Complex numerals are formed from simplex ones and base words (e.g. *shí* ‘ten’ and *bǎi* ‘hundred’) by multiplication and addition.
 - E.g. *sān-bǎi-èr-shí-yī* ‘three-hundred-two-ten-one’ (3 * 100 + 2 * 10 + 1 = 321)

I. *Jǐ* is a Simplex Numeral:

- Jǐ* can occur before base words by way of multiplication.
- Jǐ* can occur as the last digit in a complex numeral.

- (3) a. *jǐ-shí* ‘several-ten’ cf. *èr-shí* ‘two-ten’
b. *èr-shí-jǐ* ‘two-ten-several’ cf. *èr-shí-yī* ‘two-ten-one’

- Jǐ* could not replace a complex numeral embedded in a more complex one.

- (4) a. *sān-bǎi-èr-shí-yī* ⇒ **sān-bǎi-jǐ* Intended: 301 – 399
b. *sān-bǎi-líng-yī* ⇒ *sān-bǎi-líng-jǐ* ~ 301 – 309
☞ *líng* ‘zero’ is used to occupy (sequences of) digits that are zero in number.

II. *Jǐ* in questions: using *jǐ* in questions is infelicitous if the number inquired about is conventionally known to be larger than 10.

- (5) *nǐ-jǐ-suì?* ‘you-how.many-years.old’ (How old are you?)
☞ Asking a child ✓ vs. Asking an adult ✗
cf. *nǐ-duōshǎo-suì?* ☞ Asking anyone ✓
☞ *duōshǎo* ‘how.many’ is another question word for numbers but has no domain restriction.

Q: But why could the question in (2) be replied by numbers larger than 10?
A: Because a (sincere) question on number typically requires the speaker to be ignorant about the exact range of the answer, which contradicts *jǐ*’s clear-cut lexical domain restriction and enlarges the domain pragmatically.

III. *Jǐ* in declaratives: When used as existentials *jǐ* normally cannot cover numbers larger than 9.

- (6) *Zhāngsān* borrowed *jǐ* books. ⇒ *Zhāngsān* borrowed books ≥ 10

Q: But why does *jǐ* seem to not cover 1 either in this case?

A: Because it competes with *yī* ‘1’ or $\emptyset$, which are default expressions to express 1 and thereby make the domain of *jǐ* shrink.

Sum: *Jǐ* is a simplex numeral with a lexically restricted domain from 1 to 9; its domain can shift due to pragmatic reasons.

Analysis (Part 2: association with different operators)

Alternative Semantics: I utilize the framework of alternative semantics, combining Hamblin (1976) and Rooth (1985, 1992) in making reference to both *wh* alternatives and focus alternatives.

- Contra Beck (2006) and Erlewine (2025), I assume that *wh* alternatives and focus alternatives are kept apart in two dimensions (cf. Li & Law 2016).

- (7) $\llbracket jǐ \rrbracket^o = \{n | 1 \leq n \leq 9\} = \text{SIMP}$, $\llbracket jǐ \rrbracket^f = \{n | n \geq 1\} = \text{NUM}$

- (8) **Reason:** *jǐ* can be associated with *only*.
Zhāngsān zhǐ jiè-le jǐ běn shū
Zhāngsān only borrow-PFV how.many/several CL book
‘Zhāngsān only borrowed several books.’ ⇒ not over 10

Existential and Question Operators: The different readings of *jǐ* come from association with existential or question operators.

- (9) a. $\llbracket \exists \text{ IP} \rrbracket_w^o = \{\exists p \in \llbracket \text{IP} \rrbracket_w^o. p(w) = 1\}$, $\llbracket \exists \text{ IP} \rrbracket_w^f = \llbracket \text{IP} \rrbracket_w^f$
b. $\llbracket \text{Q IP} \rrbracket_w^o = \llbracket \text{IP} \rrbracket_w^o$, presupposing that $|\llbracket \text{IP} \rrbracket_w^o| > 1$, $\llbracket \text{Q IP} \rrbracket_w^f = \llbracket \text{IP} \rrbracket_w^o$

- (10) **Deriving the readings of (2)**
a. LF: $[_{\text{IP}} \text{Q} / \exists [_{\text{IP}} \text{Zhāngsān borrowed } jǐ \text{ books}]]$
b. Q: $\llbracket (2) \rrbracket_w^o = \{\wedge \exists x. \text{books}(x) \wedge |x| = n \wedge \text{borrowed}(\text{ZS}, x) | n \in \text{SIMP}\}$
c. $\exists: \llbracket (2) \rrbracket_w^o = \{\wedge \exists x. \text{books}(x) \wedge |x| = n \wedge n \in \text{SIMP} \wedge \text{borrowed}(\text{ZS}, x)\}$

Analysis (Part 3: why no obligatory ignorance)

Answer: The lexical domain restriction of *jǐ* makes it *informative* in existentials, as opposed to other *wh*-items.

- After exhaustification assumed for numerals generally, *jǐ* is informative in that it would not be entailed by substitution of all of its focus alternatives.

- (11) $\neg \forall n \in \llbracket jǐ \rrbracket^f = \text{NUM}. \llbracket \phi^{j \rightarrow n} \rrbracket \Rightarrow \llbracket \phi \rrbracket$
E.g. *Zhāngsān* borrowed 20 books ⇒ *Zhāngsān* borrowed (only) *jǐ* books

- This is not the case for other *wh*-items, given that their *wh* and focus alternatives are the same.

- (12) Ordinary *wh*-items cannot associate with *only* under the existential use.
**Zhāngsān zhǐ cóng shéi nàr jiè-le běn shū*
Zhāngsān only from who/someone there borrow-PFV CL book
Intended: ‘Zhāngsān borrowed a book only from someone.’

- (13) $\llbracket shéi \rrbracket^o = \llbracket shéi \rrbracket^f = \{x | x \in D_e \wedge \text{people}(x)\}$

- The obligatory ignorance inference arises as a **Need a Reason** implicature in Lauer’s (2013) terms, i.e. to give a reason for the use of an uninformative expression.
- When *jǐ*’s environment restricts admissible focus alternatives, the ignorance effect could arise obligatorily.

- (14) *Zhāngsān jǐ diǎn zhōng lái-guò zhèr*
Zhāngsān several/how.many hour clock come-EXP here
Lit. ‘Zhāngsān came here at some hour (that I don’t know).’
☞ Alternatives of *jǐ* restricted to 1 to 12, roughly equal to SIMP

References

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